人工智能基础

作者：王五

Chapter 1: Introduction to Machine Learning

Machine Learning enables computers to learn from data without explicit programming.

Three main paradigms:

1. Supervised Learning - learns from labeled examples
2. Unsupervised Learning - finds patterns in unlabeled data
3. Reinforcement Learning - learns through trial and reward

Common applications: image recognition, spam filtering, recommendation systems.

Chapter 2: Neural Networks

Artificial Neural Networks are inspired by biological neurons.

A neuron computes: output = activation(weights * inputs + bias)

Deep Learning uses many layers to learn hierarchical representations.

Common architectures:

- CNN: image processing
- RNN/LSTM: sequential data
- Transformer: natural language processing
- GAN: generative models

Chapter 3: Practical Applications

Computer Vision:
- Image classification
- Object detection
- Semantic segmentation

Natural Language Processing:
- Sentiment analysis
- Machine translation
- Question answering
- Text generation

Tools: Python, TensorFlow, PyTorch, scikit-learn, Hugging Face Transformers.